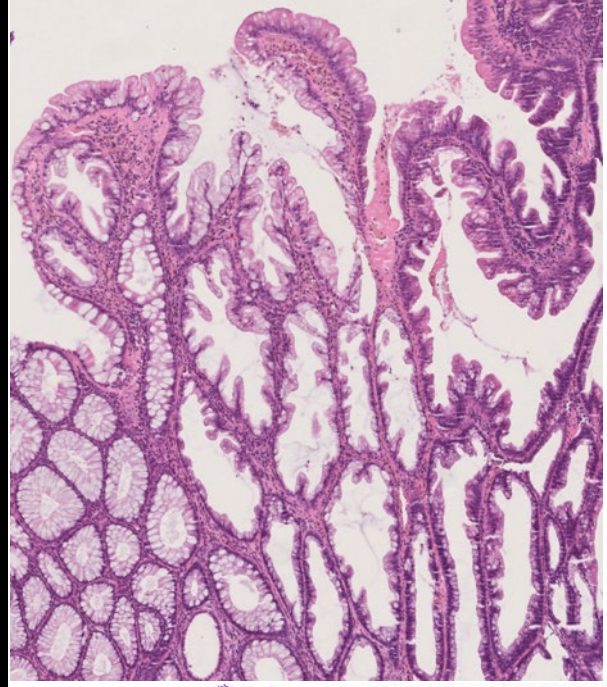
PATHOPHYSIOLOGY OF COLORECTAL CANCER

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L E A R N I N G O U T C O M E S

- Outline the pathological changes of colon polyps (non-neoplastic, conventional adenomas, and serrated lesions)
- List the proto-oncogenes and suppressor genes involved in colorectal neoplasia
- Describe the molecular pathways associated with familial forms of colorectal neoplasia

Literature link:

Robbins Basic Pathology, af Vinay Kumar, m.fl., 11th ed. Elsevier Saund, Page 520-527
Patologi, FADL'S forlag, 4.udgave, Side 301- 314
Links to the relevant papers on moodle page (included in pensum)

O U T L I N E

- Non-neoplastic polyps
  - Inflammatory polyps
  - Hamartomatous polyps
- Epithelial polyps
  - Conventional adenomas
  - Serrated lesions
- Hereditary Colorectal cancer forms

POLYP CLASSIFICATION

Robbins classification

Non-neoplastic polyps	Neoplastic polyps
Inflammatory	Adenoma
Hamartomatous	
Hyperplastic	

→ Based on malignant potential
Useful for understanding cancer risk
Simplified

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- Hamartomatous
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Neoplastic polyps

- Adenoma

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WHO 2019 /Current pathology classification

Conventional adenomas

- Tubular
- Villous
- Tubulovillous

Serrated lesions

- Hyperplastic polyp (HP)
- Sessile serrated lesion (SSL)
- Traditional serrated adenoma (TSA)

Based on morphology and molecular features
Current diagnostic pathology guidelines

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O U T L I N E

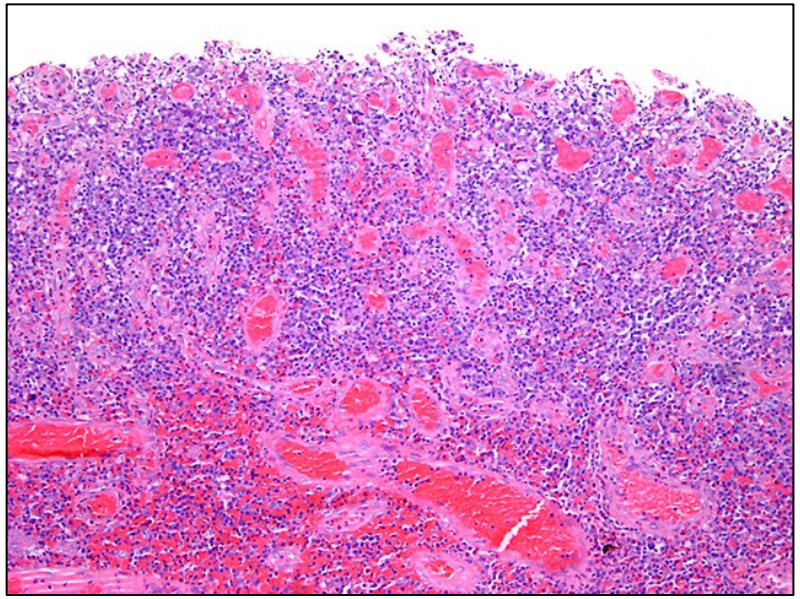
- **Non-neoplastic polyps**
  - Inflammatory polyps
  - Hamartomatous polyps
    - Juvenile polyps
    - Peutz-Jeghers polyp

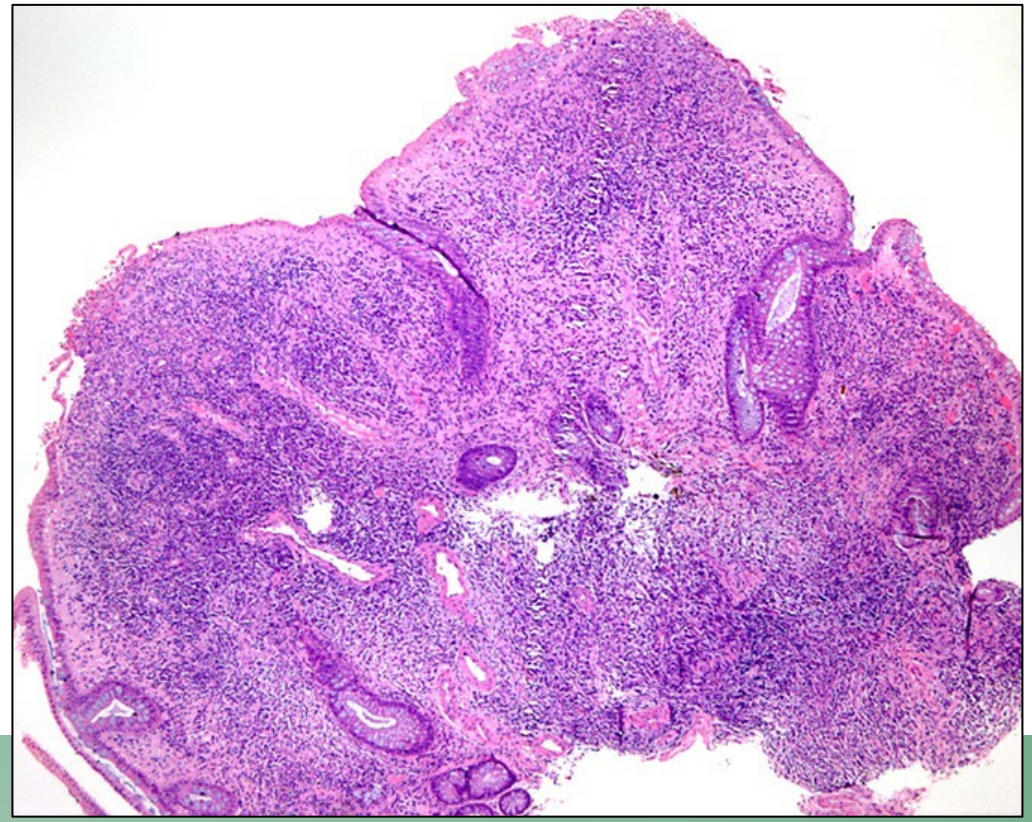
- Epithelial polyps

- Hereditary Colorectal cancer forms

INFLAMMATORY POLYPS

- *General*
  - Non-neoplastic polyps arising from repeated cycles of mucosal injury and healing
  - Often associated with inflammatory bowel disease, anastomosis, ischemic colitis or infection
- *Gross morphology*
  - Typically Sessile (<3 cm)
  - Pedunculated or filiform
- *Histopathologic classification*
  - Expanded lamina propria
  - Crypt abscesses or cryptitis
  - Surface erosion and ulceration
  - Granulation tissue (thin walled and dilated vessels surrounded by mixed neutrophilic and lymphoplasmacytic inflammation
  - Active and chronic inflammation
- *Clinical significance*
  - Marker of previous mucosal injury
  - No intrinsic malignant potential



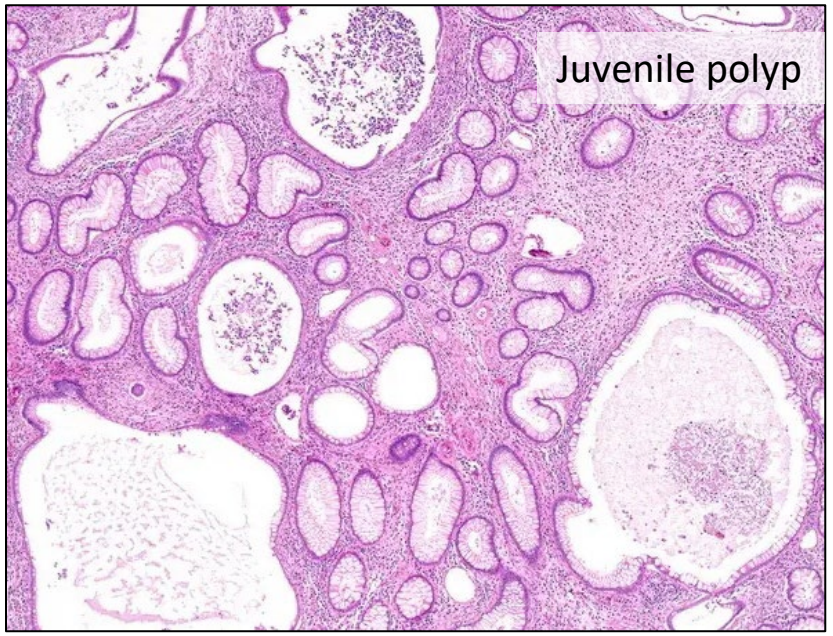


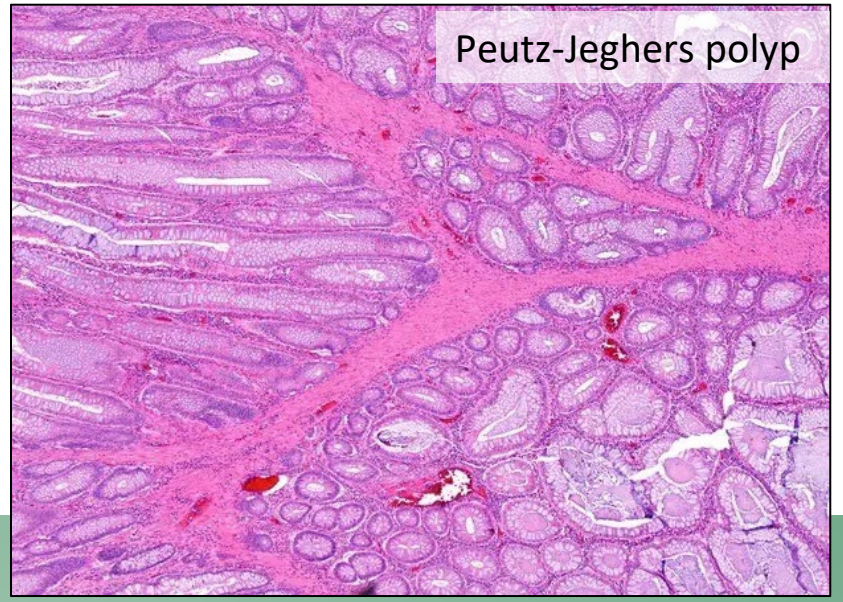
HAMARTOMATOUS POLYPS

- *General*
  - Non-neoplastic polyps composed of a disorganized overgrowth of tissues normally present at the site
  - Characterized by abnormal mixture of epithelial and stromal components
  - May occur sporadically or as part of an inherited syndrome
- *Major subtypes*
  - Juvenile polyp (*cystically dilated glands*)
  - Peutz-Jeghers polyp (*arborizing smooth muscle*)

Solitary hamartomatous polyps are usually benign.

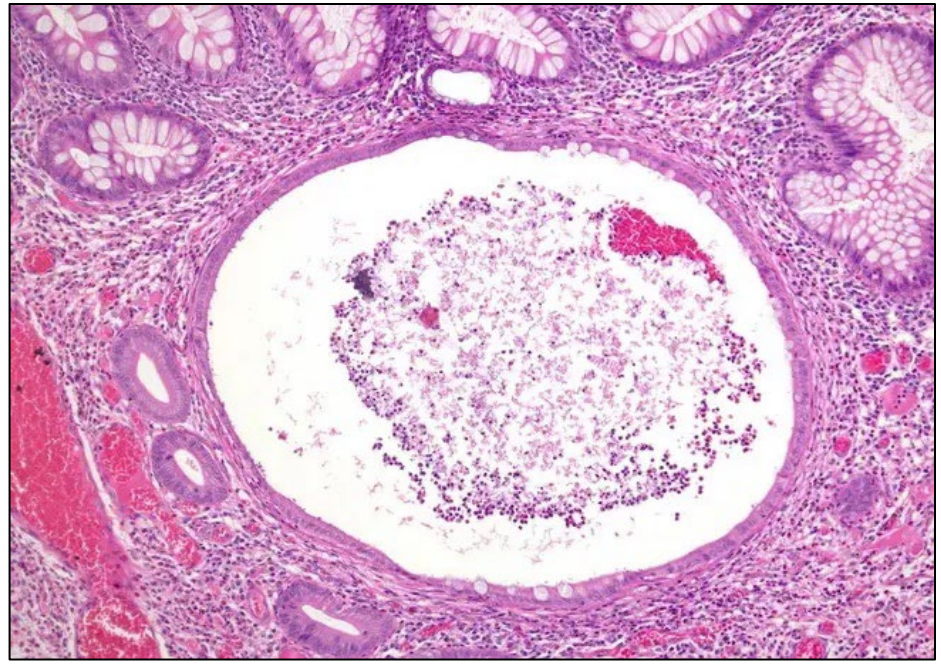
In contrast, hamartomatous polyposis syndromes carry an increased risk of gastrointestinal and extraintestinal malignancies.

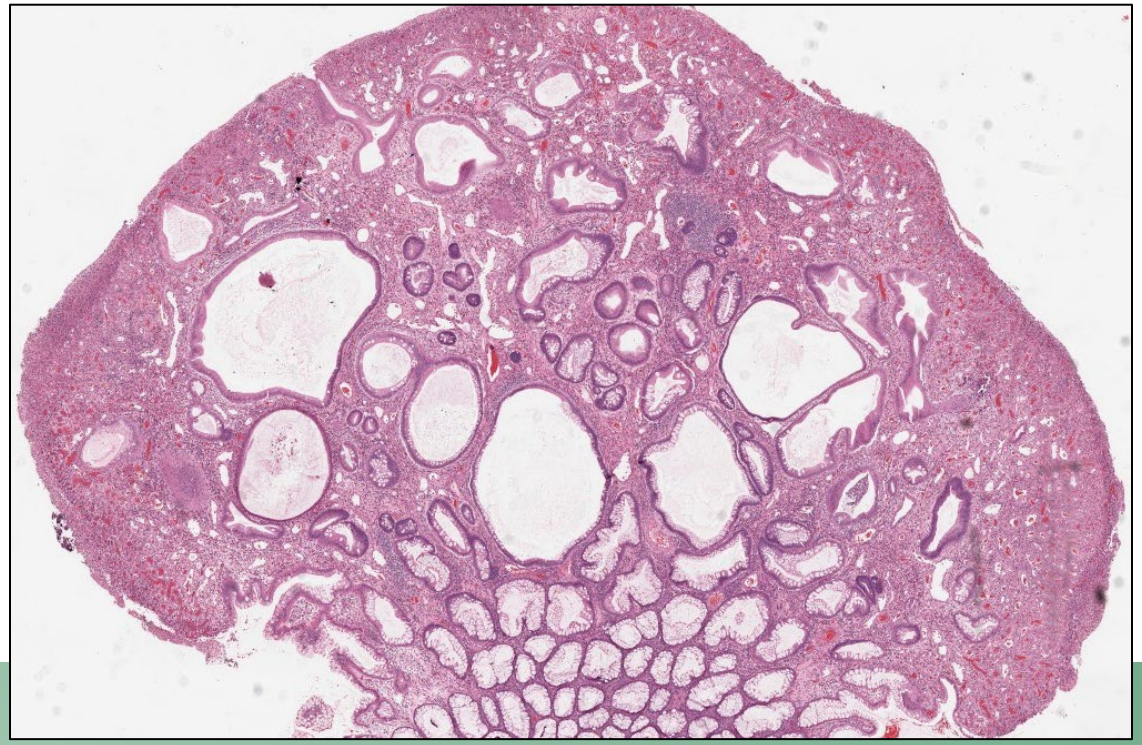




JUVENILE POLYPS

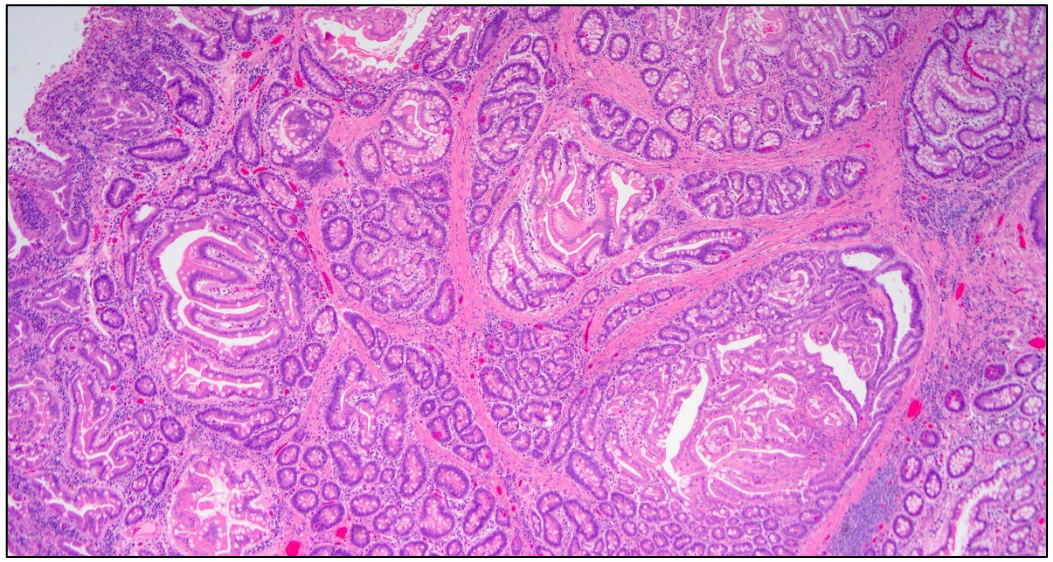
- *General*
  - Most common hamartomatous polyp in children (typically 3-10 years of age)
  - Most commonly located in rectosigmoid colon
- *Histological features*
  - Cystically dilated mucus filled glands
  - Dilated glands contain abundant mucin and inflammatory debris
  - Edematous and inflamed stroma
  - Surface shows erosion, ulceration or granulation tissue
- *Juvenile Polyposis Syndrome*
  - Multiple juvenile polyps
  - Germline mutation in *SMAD4* or *BMPR1A*
  - Autosomal dominant inheritance
  - Increased risk of colorectal and gastric carcinoma

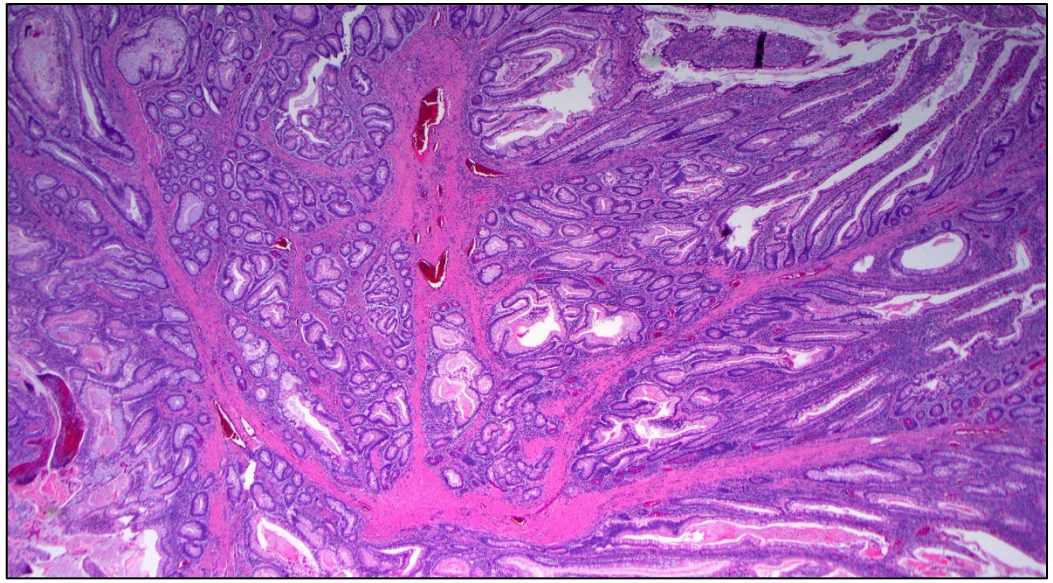




PEUTZ-JEGHERS POLYPS

- *General*
  - Most commonly located in the small intestine
  - May also occur in colon, rectum, and stomach
  - Sporadic polyps are extremely rare
- *Histological features*
  - Arborizing ("tree-like") smooth muscle bundles extending from the muscularis mucosae
  - Smooth muscle divides the polyp into complex branching lobules
  - Glands are lined by non-dysplastic intestinal epithelium
  - Variable cystic dilatation may be present
- *Peutz-Jeghers Syndrome*
  - Multiple Peutz-Jeghers polyps (at least 3)
  - Germline mutation in *STK11* (formerly called *LKB1*)
  - Autosomal dominant inheritance
  - Increased risk of gastrointestinal carcinoma, pancreatic carcinoma, breast carcinoma, gynecologic malignancies





O U T L I N E

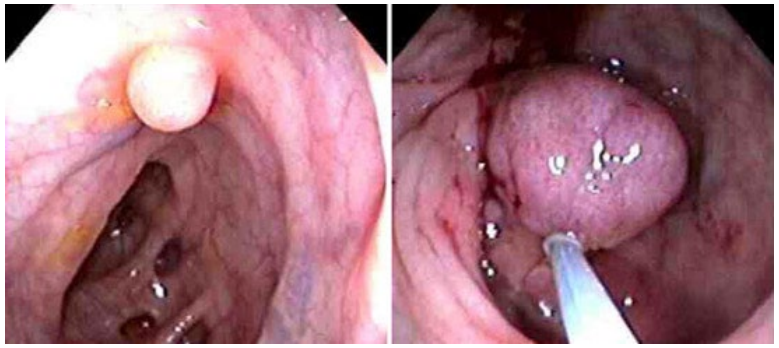
- Non-neoplastic polyps

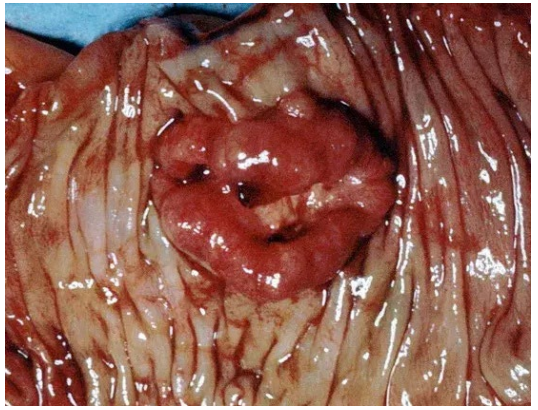
- **Epithelial polyps**
  - **Conventional adenomas**
    - Tubular adenoma
    - Villous adenoma
    - Tubulovillous adenoma
  - Serrated lesions
    - Hyperplastic polyps
    - Sessile serrated lesion
    - Traditional serrated adenoma

- Hereditary Colorectal cancer forms

CONVENTIONAL ADENOMA

- *Definition*
  - Benign epithelial neoplasms composed of dysplastic glandular epithelium
  - Confined above lamina muscularis mucosae
  - Precursor lesion of CRC (conventional pathway)





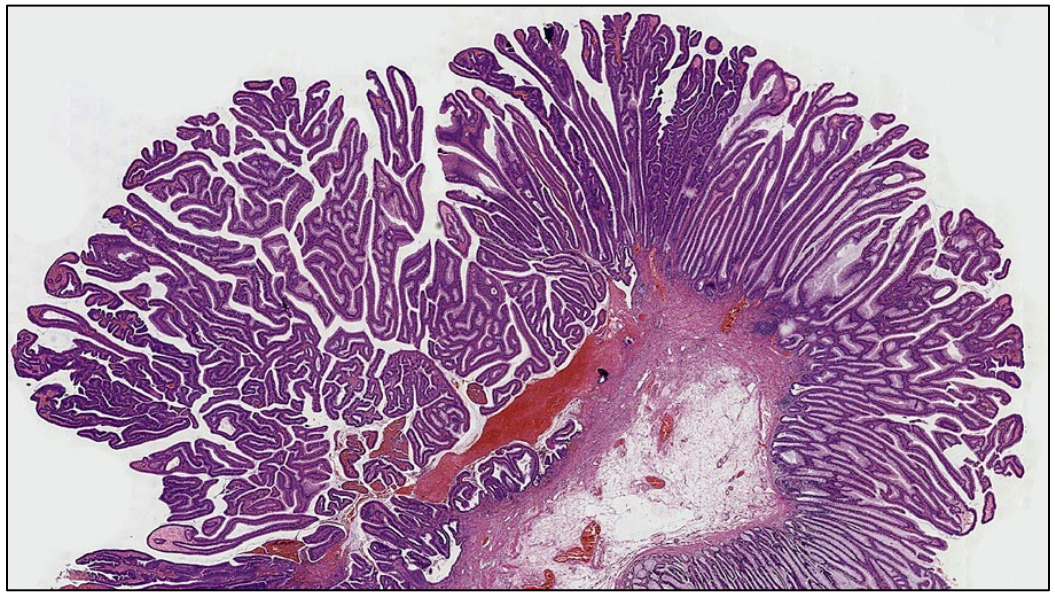
- *Gross morphology*
  - Sessile
  - Pedunculated
  - Smooth, velvety or lobulated surface
  - Variable size

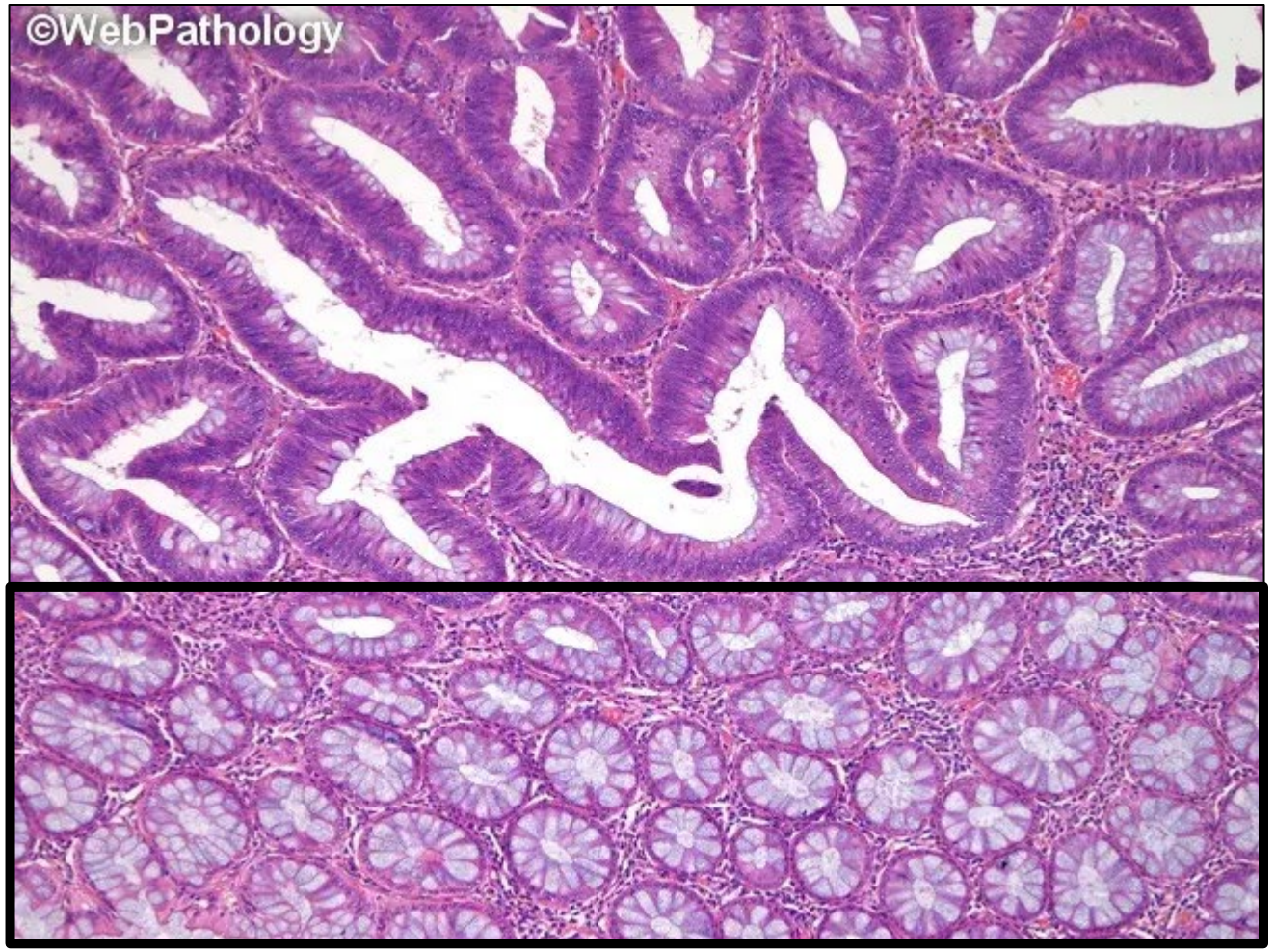
- *Histopathologic classification*
  - gland architecture
    - Tubular (<25% villous component)
    - Tubulovillous (25-75% villous component)
    - Villous (>75% villous component)
  - cytologic atypia and architectural complexity
    - Low-grade neoplasia
    - High-grade neoplasia

In the literature, the terms *dysplasia* and *neoplasia* are often used interchangeably

ADENOMA: GENERAL HISTOPATHOLOGY

- *Architectural atypia*
  - Gland crowding
  - Gland branching
  - Villous architecture (finger-like)

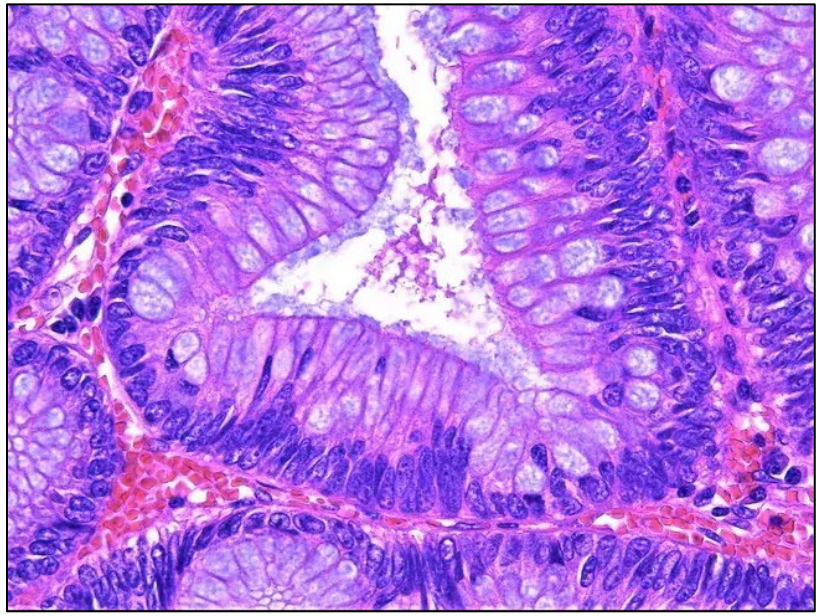


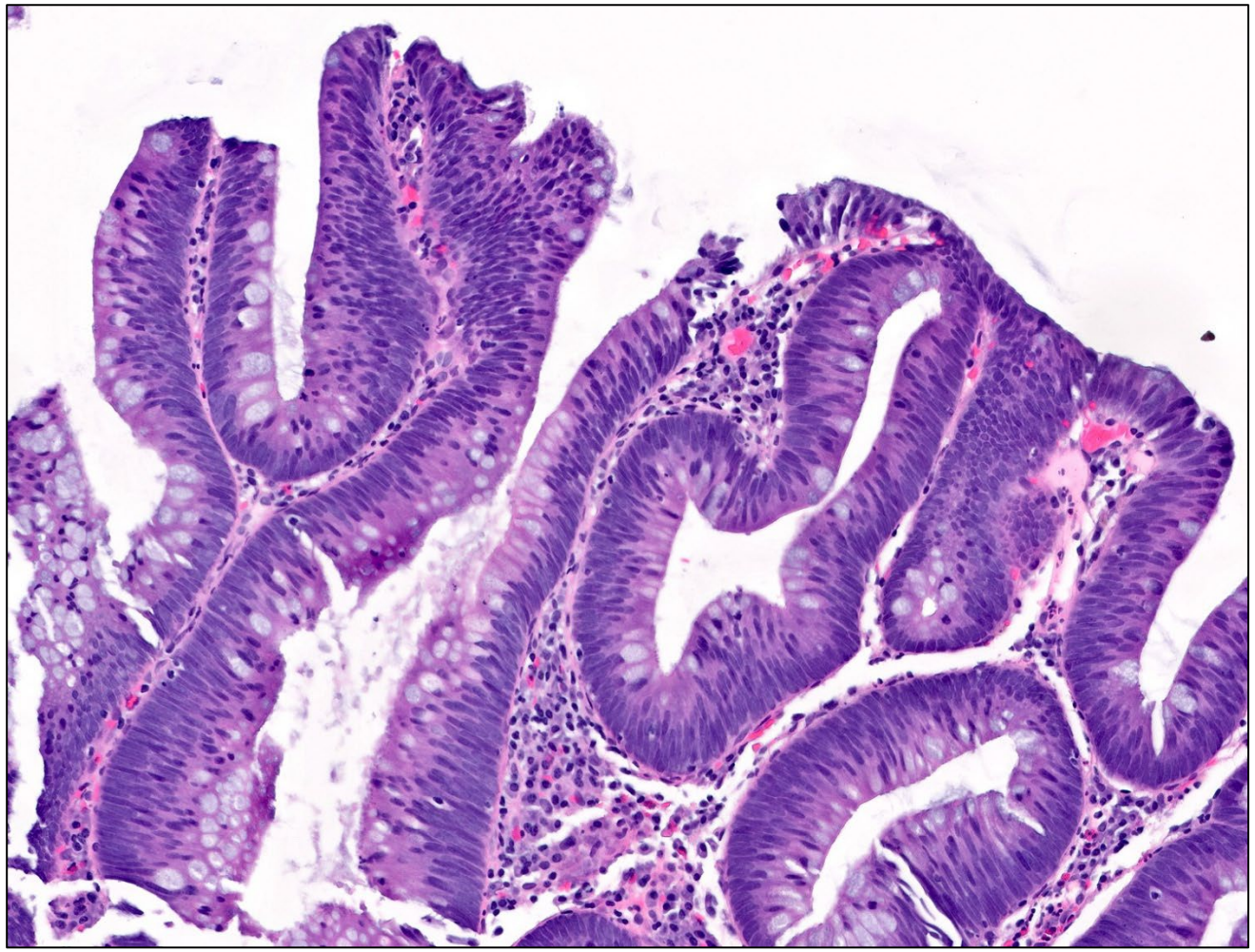


Normal colonic mucosa

ADENOMA: GENERAL HISTOPATHOLOGY

- *Cytological atypia*
  - Hypercellularity
  - Nuclear pseudostratification
  - Loss of polarity
  - Pleomorphy
  - Atypical mitoses
  - Loss of goblet cells



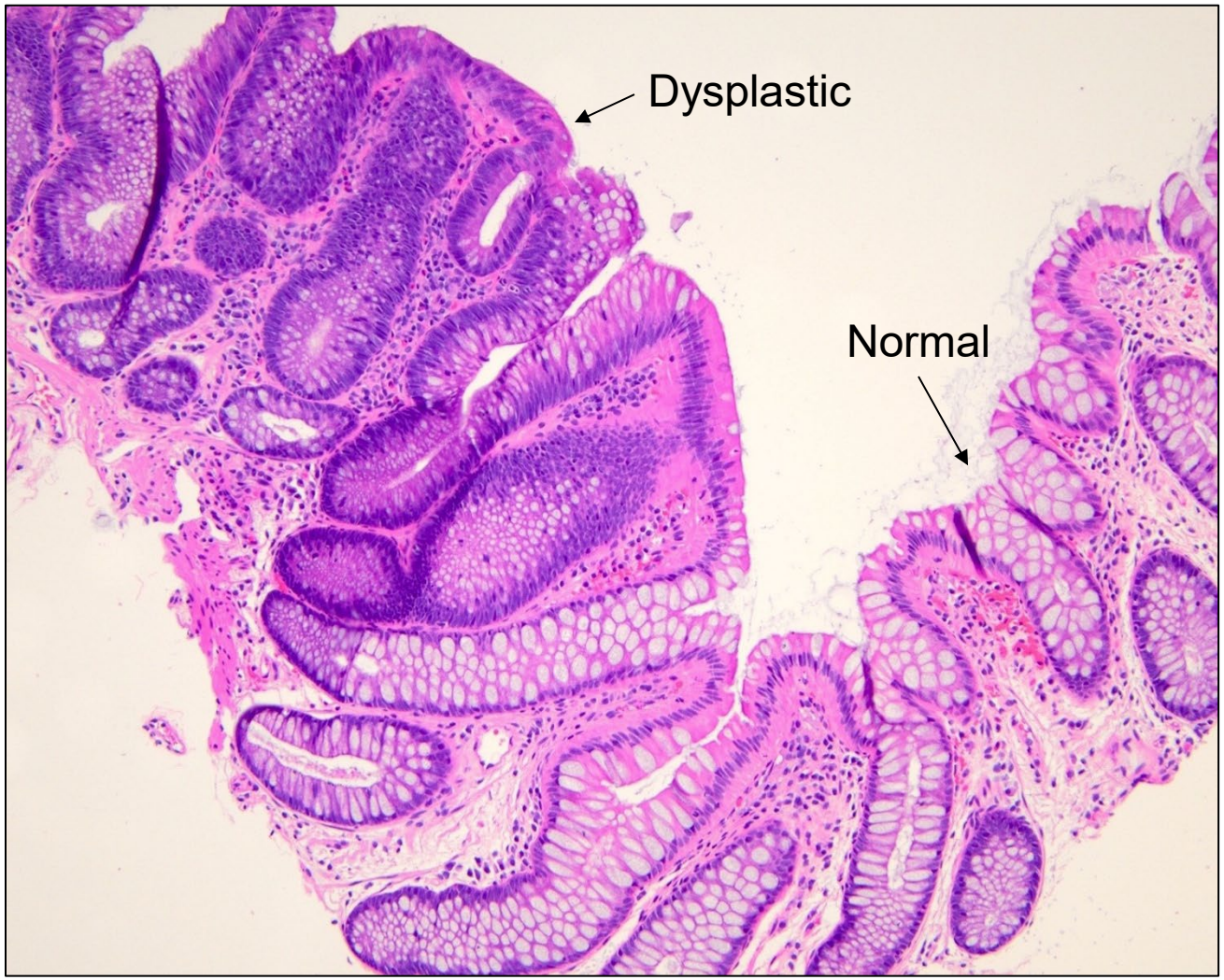


ADENOMA: GENERAL HISTOPATHOLOGY

- *Architectural atypia*
  - Gland crowding
  - Gland branching
  - Villous architecture

- *Cytological atypia*
  - Hypercellularity
  - Hypercromasia (basophilic appearance)
  - Nuclear pseudostratification
  - Loss of polarity
  - Pleomorphy
  - Atypical mitoses
  - Loss of goblet cells

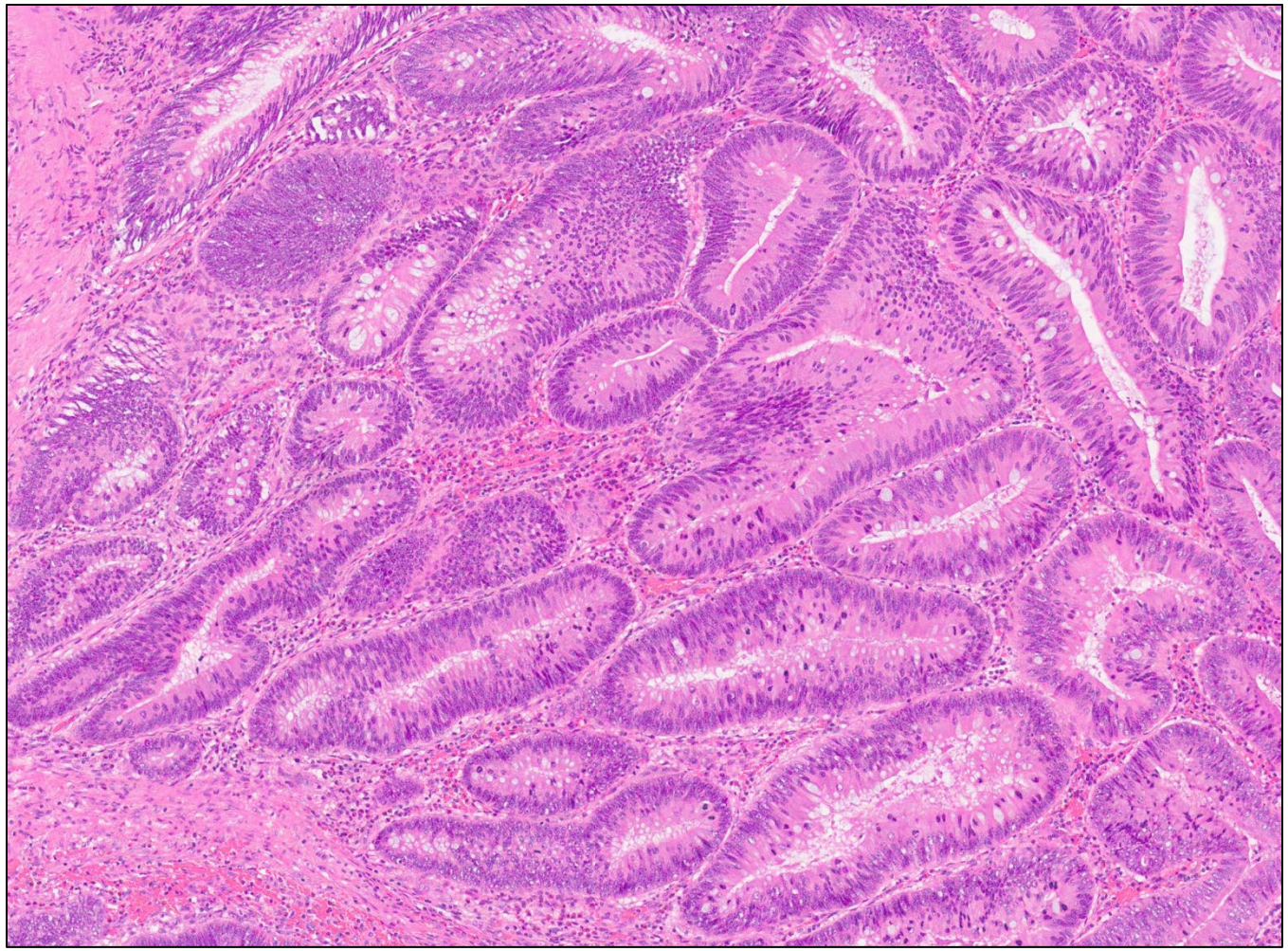
Adenomas are composed of dysplastic epithelium



ADENOMA: LOW-GRADE NEOPLASIA

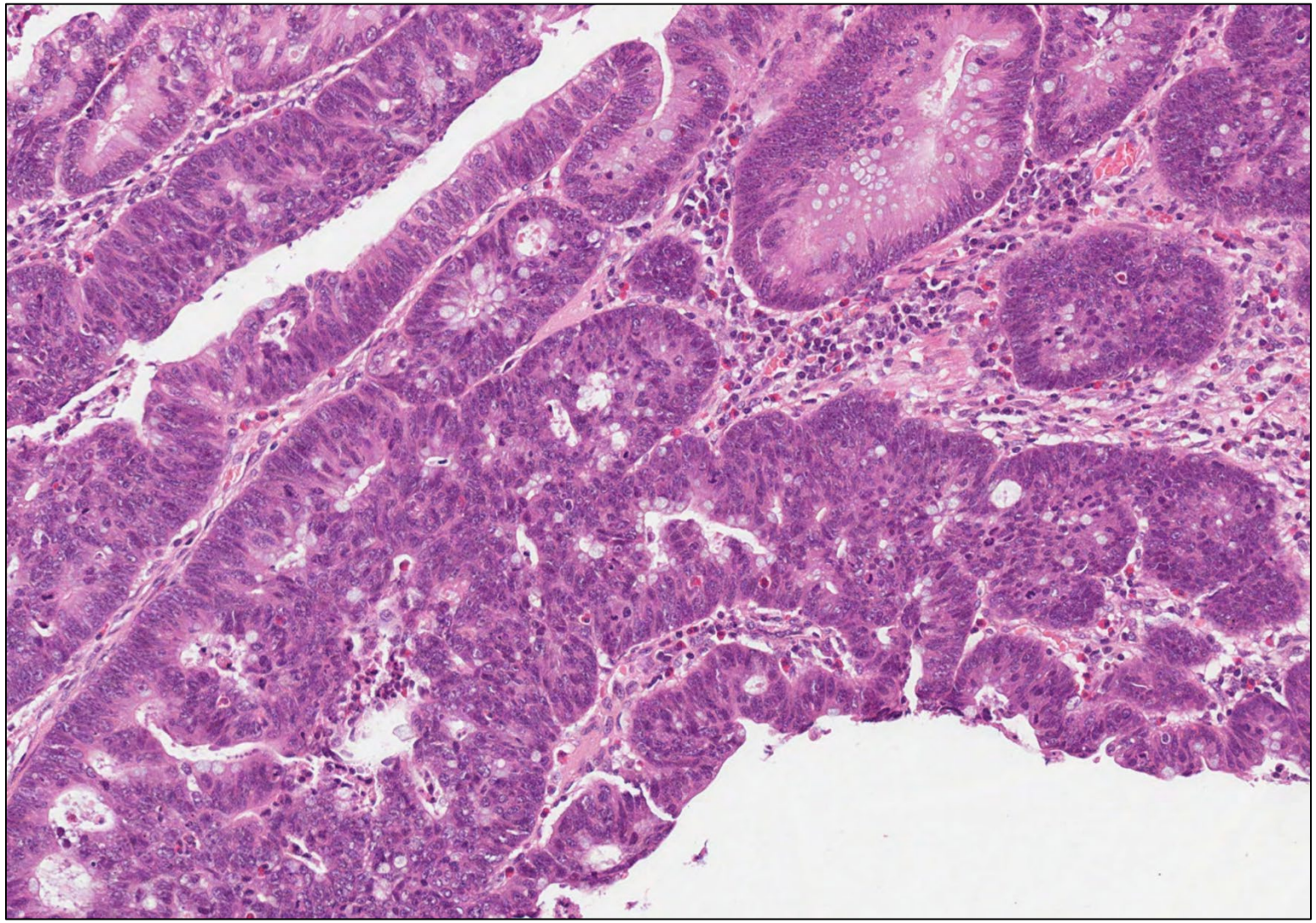
- Histological features
  - Simple adenomatous glands
  - Occasional gland branching
  - No cribriform architecture
  - Mild loss of polarity
  - Mild pleomorphism
  - Pseudostratified nuclei confined to the lower 2/3
  - Rare atypical mitoses



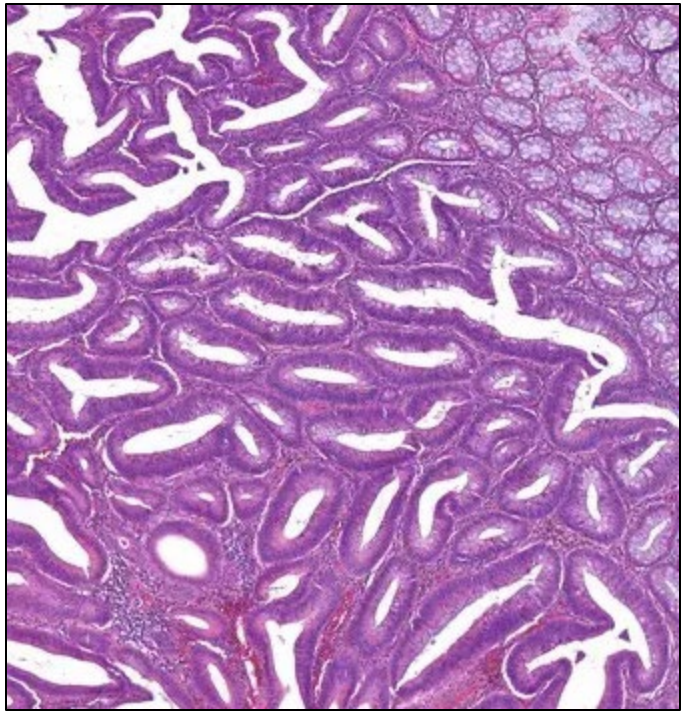
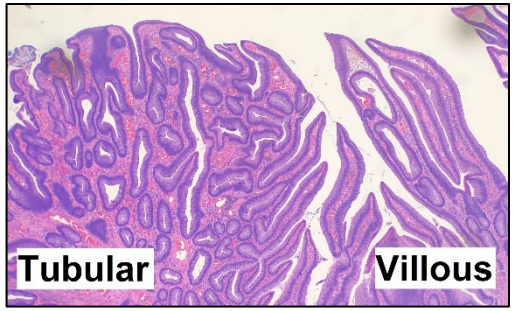
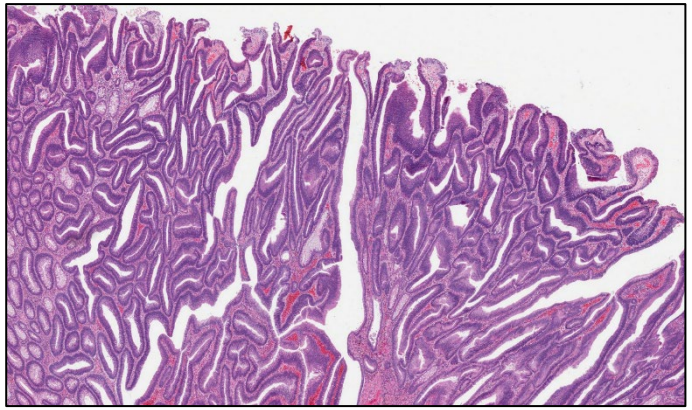
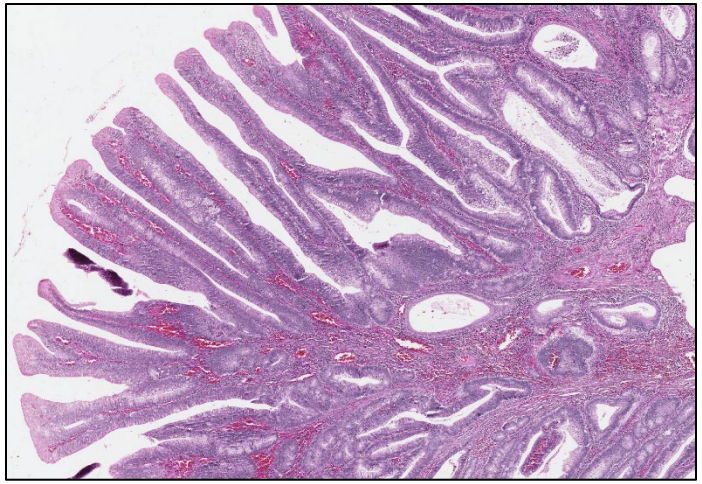


ADENOMA: HIGH-GRADE NEOPLASIA

- *Histological features*
  - Closely packed and complex glands
  - Frequent glandular angulation
  - Back-to-back glandular growth pattern
  - Cribriform architecture with intraluminal debris
  - Prominent nucleoli
  - Loss of cellular polarity
  - Enlarged, pleomorphic nuclei
  - Frequent atypical mitoses



CONVENTIONAL ADENOMA

	Tubular	Tubulovillous	Villous
Gross appearance	Often pedunculated	Sessile or pedunculated	Often sessile, velvety surface
Histology	Rounded tubular glands	mixture of tubular and villous structures	Slender villous (finger-like) projections
Architecture	<25% villous component	25-75% villous component	>75% villous component
Frequency	Most common (~70% of adenomas)	Intermediate	<5% of adenomas
Microscopic appearance		 	

CONVENTIONAL ADENOMA

- Risk of malignant progression
  - Increases with
    - Larger size
    - High-grade neoplasia

Although villous adenomas are often large and frequently exhinbit high-grade neoplasia, **villous architecture is no longer considered an independent risk factor.**

- Precursor lesions of the conventional CRC pathway

CRC PATHOGENESIS: CONVENTIONAL ADENOMA-CARCINOMA SEQUENCE

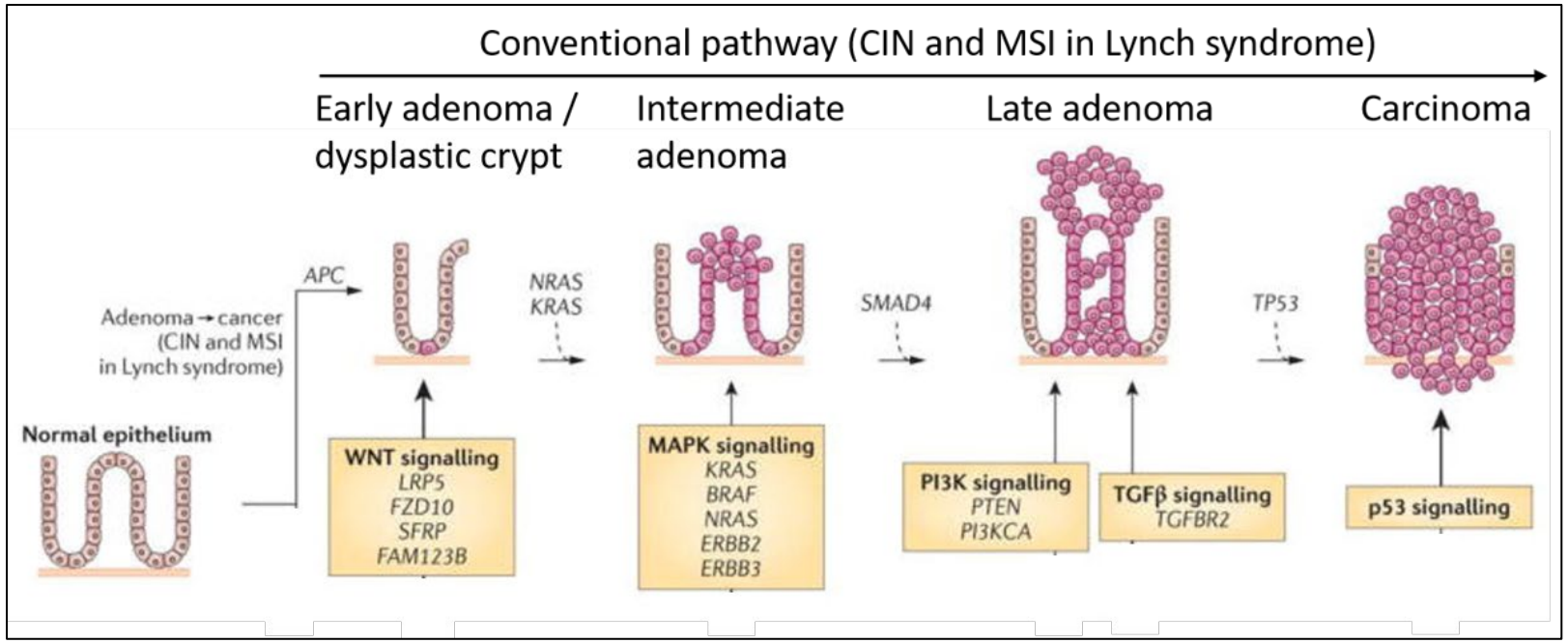
- Conventional pathway → multistep progression from adenoma to carcinoma
  - Driven by accumulating genetic and/or epigenetic aberrations
  - First described by Fearon and Vogelstein in 1990
  - Accounts for ~70% of sporadic CRCs

- Driver mutations
  - *APC* (early event)
  - *TP53* (late event)

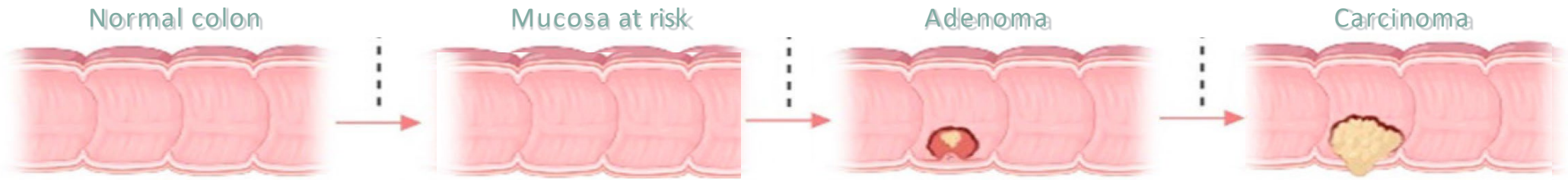
- Other molecular alterations
  - *KRAS* activation
  - Loss of SMAD4/TGF-β signaling

- Sporadic CRC
  - Somatic mutations
  - Chromosomal instability (CIN)

CIN: accumulation of numerical and structural chromosomal abnormalities



COLON CARCINOGENESIS-ADENOMA-CARCINOMA SEQUENCE



First hit: Germline (inherited) or somatic (acquired) mutation

APC at 5q21

Second hit: Loss of the remaining *APC* allel

APC loss

LOH: loss of heterozygosity

Protooncogenes mutation

K-RAS at 12p12

Loss of additional tumor suppressor genes

TP53 at 17p13
LOH at 18q21 (*SMAD2 and 4*)

Additional mutations
Chromosomal instability (CIN)

Additional mutations

Knudson's Two-Hit hypothesis: both alleles of a tumor suppressor gene must be inactivated

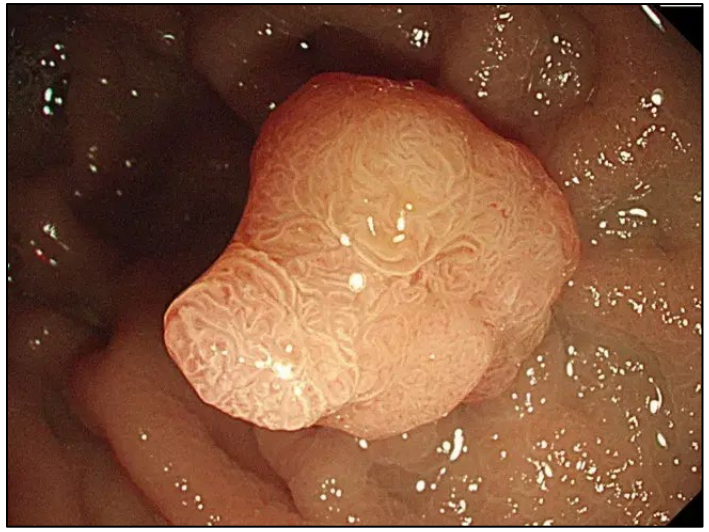
O U T L I N E

- Non-neoplastic polyps

- **Epithelial polyps**
  - Conventional adenomas
    - Tubular adenoma
    - Villous adenoma
    - Tubulovillous adenoma
  - **Serrated lesions**
    - Hyperplastic polyps
    - Sessile serrated lesion
    - Traditional serrated adenoma

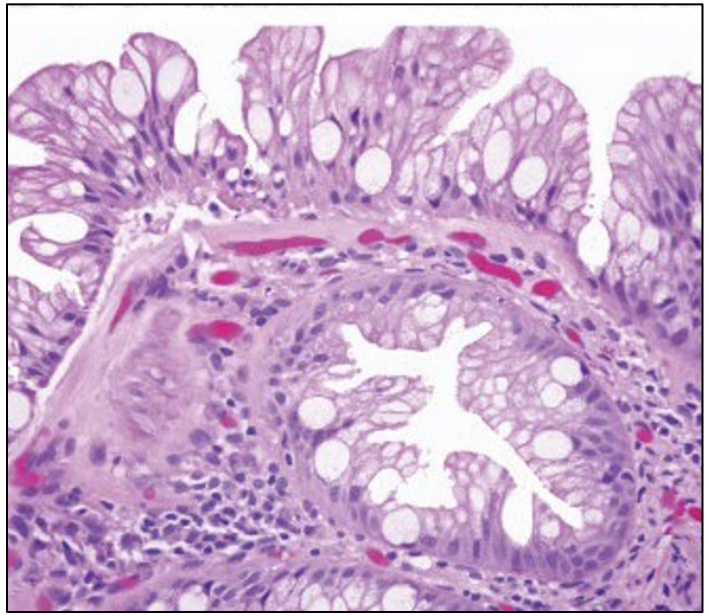
- Hereditary Colorectal cancer forms

SERRATED LESIONS



- *General*
  - Colorectal epithelial lesions characterized by a serrated ("sawtoothed" or stellate) crypt architecture
- *Gross morphology*
  - Sessile
  - Pedunculated
- *Histopathologic classification*
  - crypt architecture
    - Hyperplastic polyp (HP)
    - Sessile serrated lesion (SSL)
    - Traditional serrated adenoma (TSA)
  - In SSL, dysplasia is reported as absent or present (SSLD)
  - In TSA, grading as low-grade or high-grade neoplasia is performed only when areas of conventional adenomatous dysplasia are present.
- *Clinical significance*
  - SSL and TSA are precursor lesions of CRC (serrated pathway)
  - Hyperplastic polyps have little or no malignant potential

SSL was previously termed sessile serrated adenoma/polyp (SSA/P). This term is still encountered in older literature.



HYPERPLASTIC POLYP (HP)

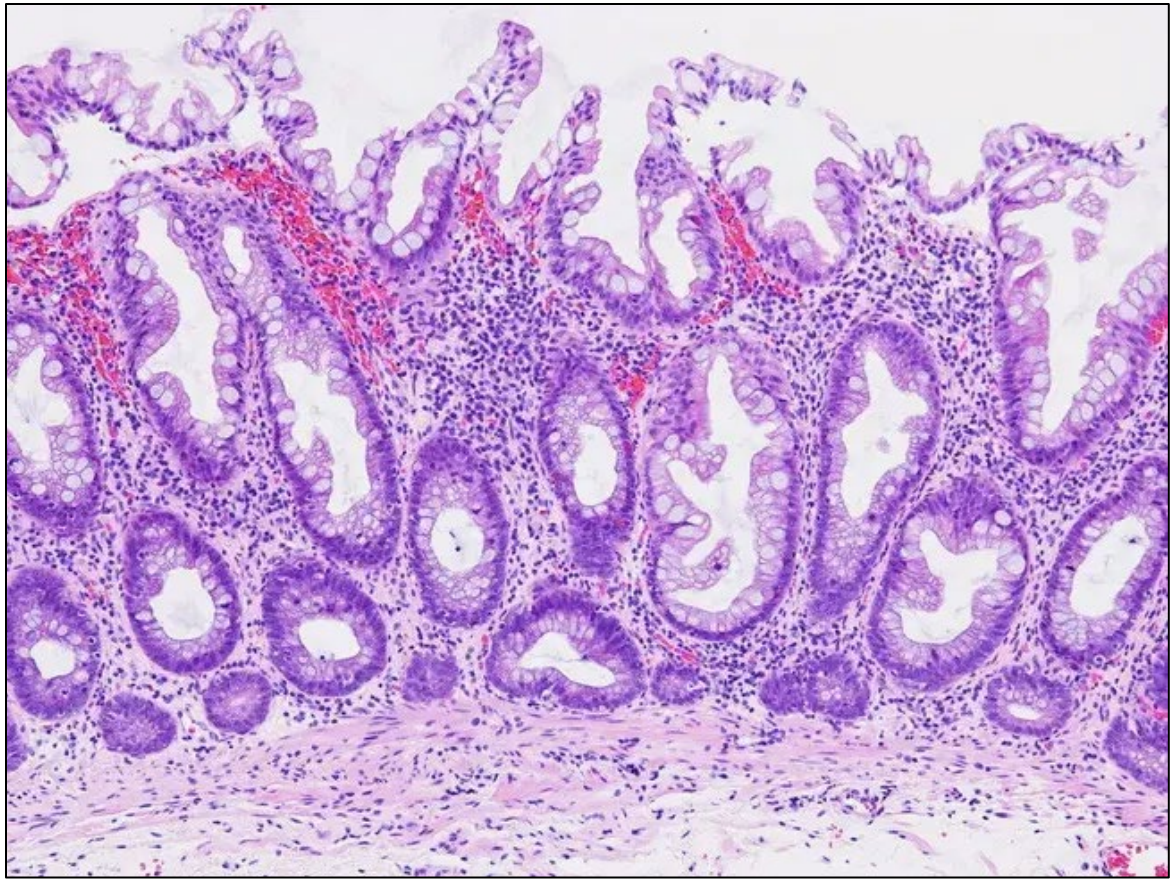
❖ *General*

- Most common serrated colorectal polyp (~ 75%)
- Predominantly located in distal colon (particularly sigmoideum) and rectum
- Often present as multiple polyps, particularly in the rectum
- Usually smaller than 5 mm

❖ *Histological features*

- Serrated architecture with a sawtooth appearance
  serration is mostly confined to the upper third of the crypts
- No crypt dilatation, branching, or horizontal growth
- Surface epithelial tufting
- No dysplasia or cytologic atypia
- Mitotic activity is limited to the base of the crypts

Main differential diagnosis:
Sessile serrated lesion (SSL)

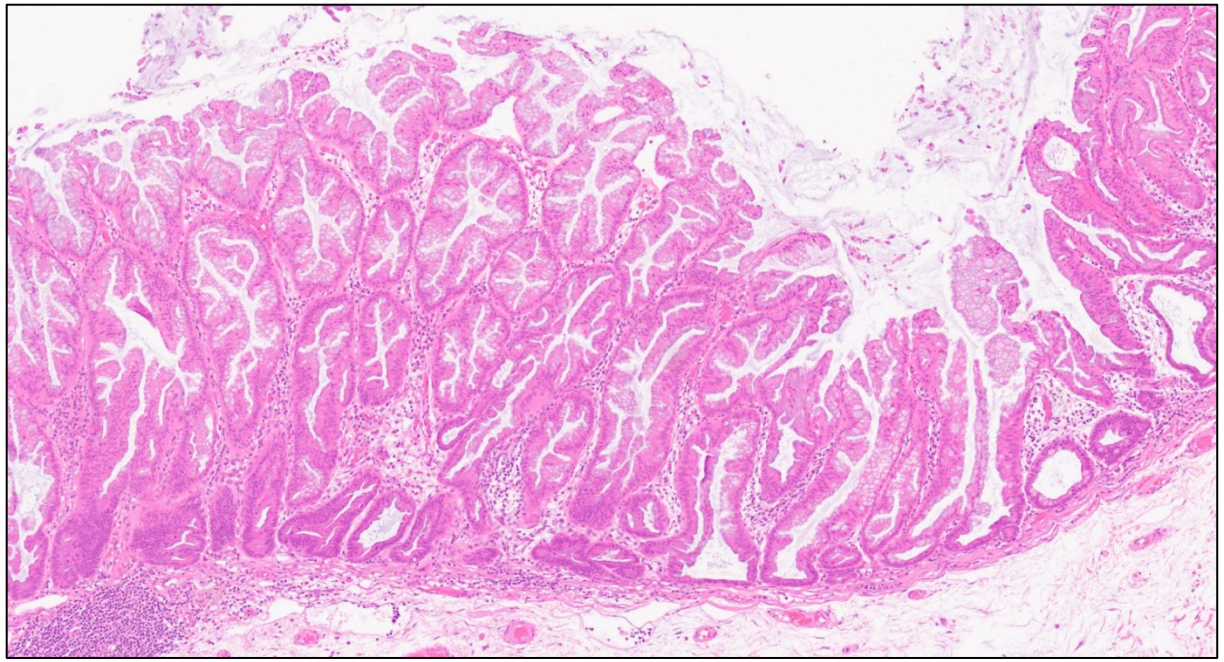


SESSILE SERRATED LESION(SSL)

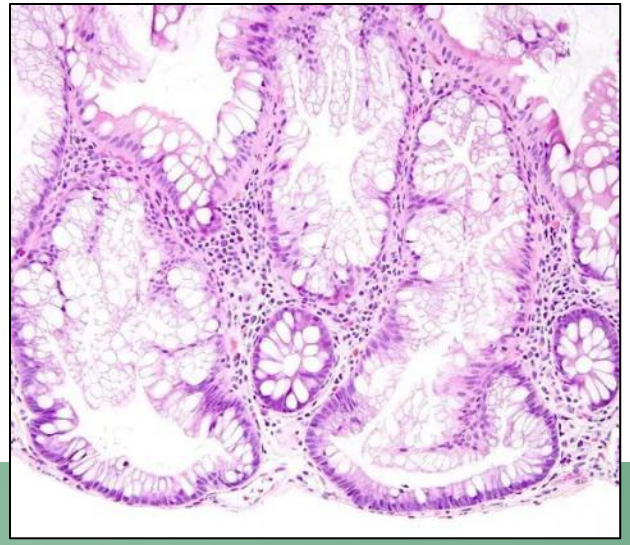
- *General*
  - Accounts for ~25% of serrated colorectal polyps
  - Predominantly located in proximal (right-sided) colon

- *Histological features*
  - Serrated architecture with a sawtooth appearance
    serration extends to the crypt base
  - Structural crypt abnormalities
    basal crypt dilation
    branching
    horizontal growth
  - Crypt bases are often L-shaped, T-shaped, or boot-shaped

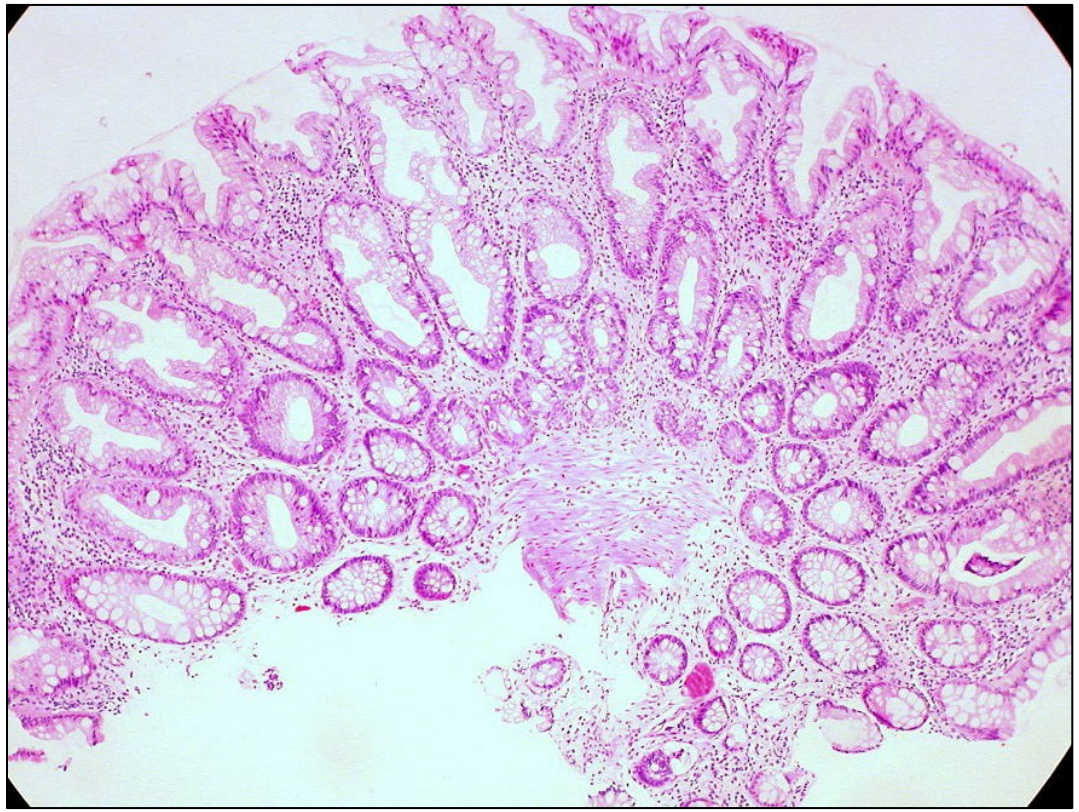
One unequivocal characteristic crypt is sufficient for SSL diagnosis







HYPERPLASTIC POLYP (HP)



- Symmetric vertical growth
- Straight, narrow crypts
- Serration limited to the upper third of the crypts
- **Crypt bases are narrow and non-dilated**

SESSILE SERRATED LESION (SSL)



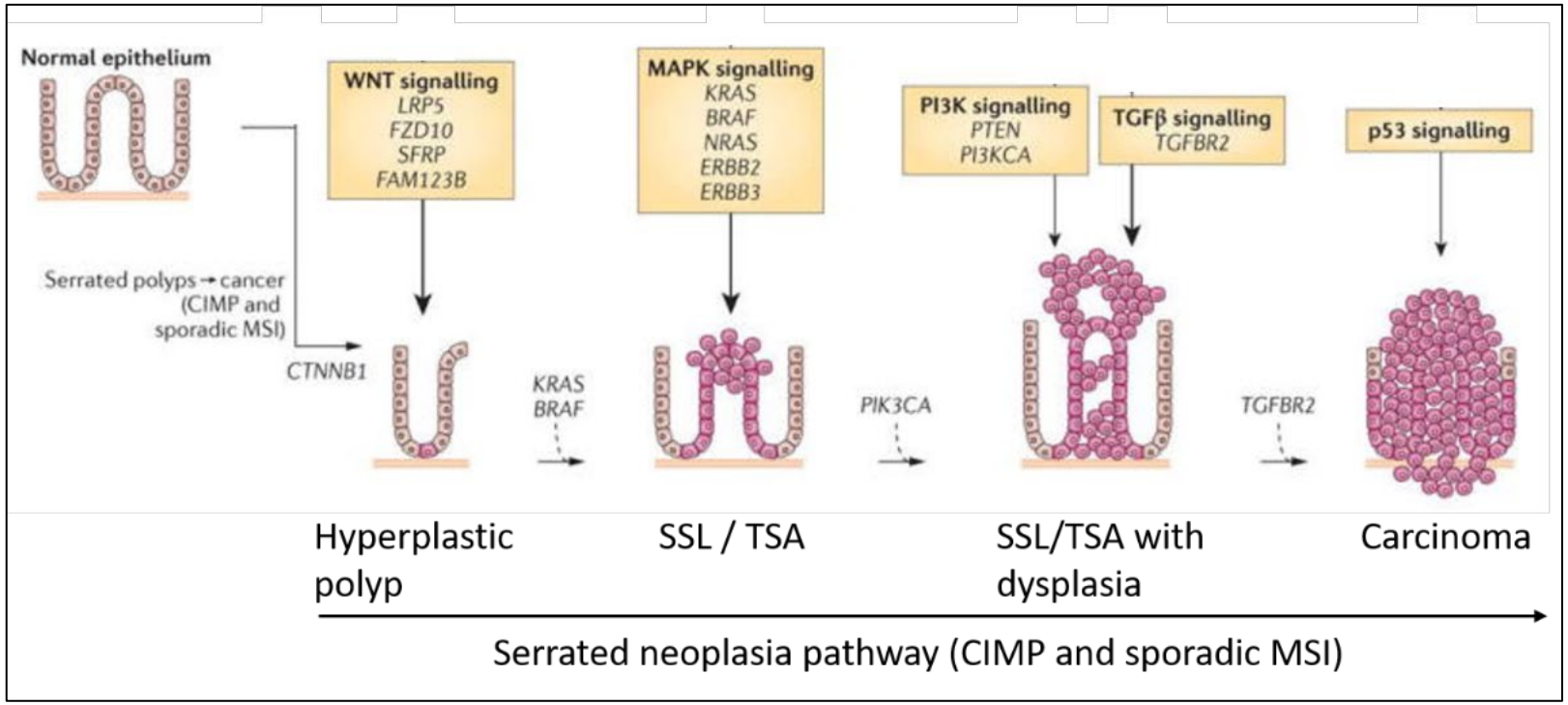
- Asymmetric horisontal growth along muscularis mucosae
- Serration extends to the crypt base
- **Crypt bases are dilated and L-, T-, or boot-shaped**

CRC PATHOGENESIS: SERRATED PATHWAY

- Epigenetic alterations play a significant role in pathogenesis

- Driver mutations
  - *BRAF* (most common, up to 50%)
  - *KRAS* (smaller subset)

- Key molecular events
  - CpG island methylation (CIMP)
  - Microsatellite instability (MSI)



O U T L I N E

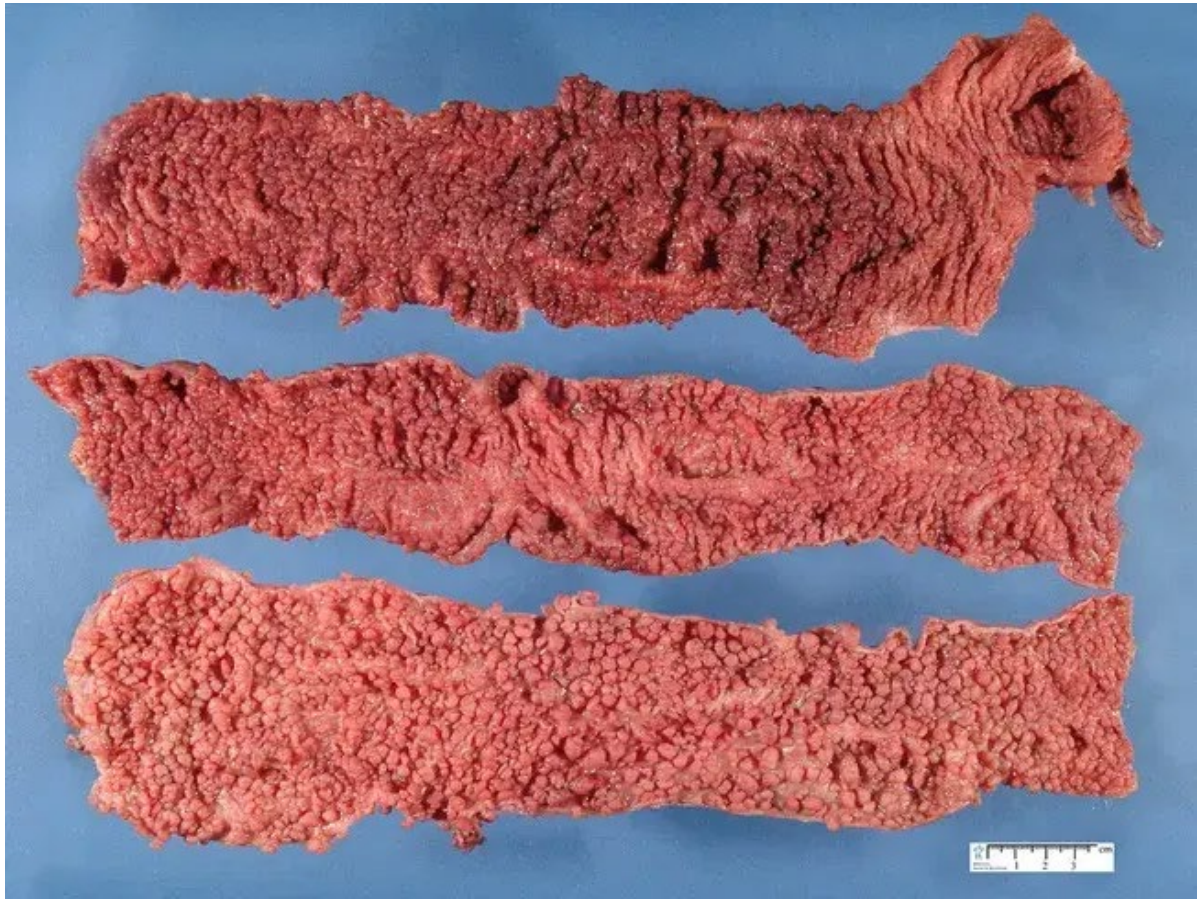
- Non-neoplastic polyps

- Epithelial polyps

- **Hereditary Colorectal cancer forms**
  - Familial adenomatous polyposis
  - Lynch syndrome

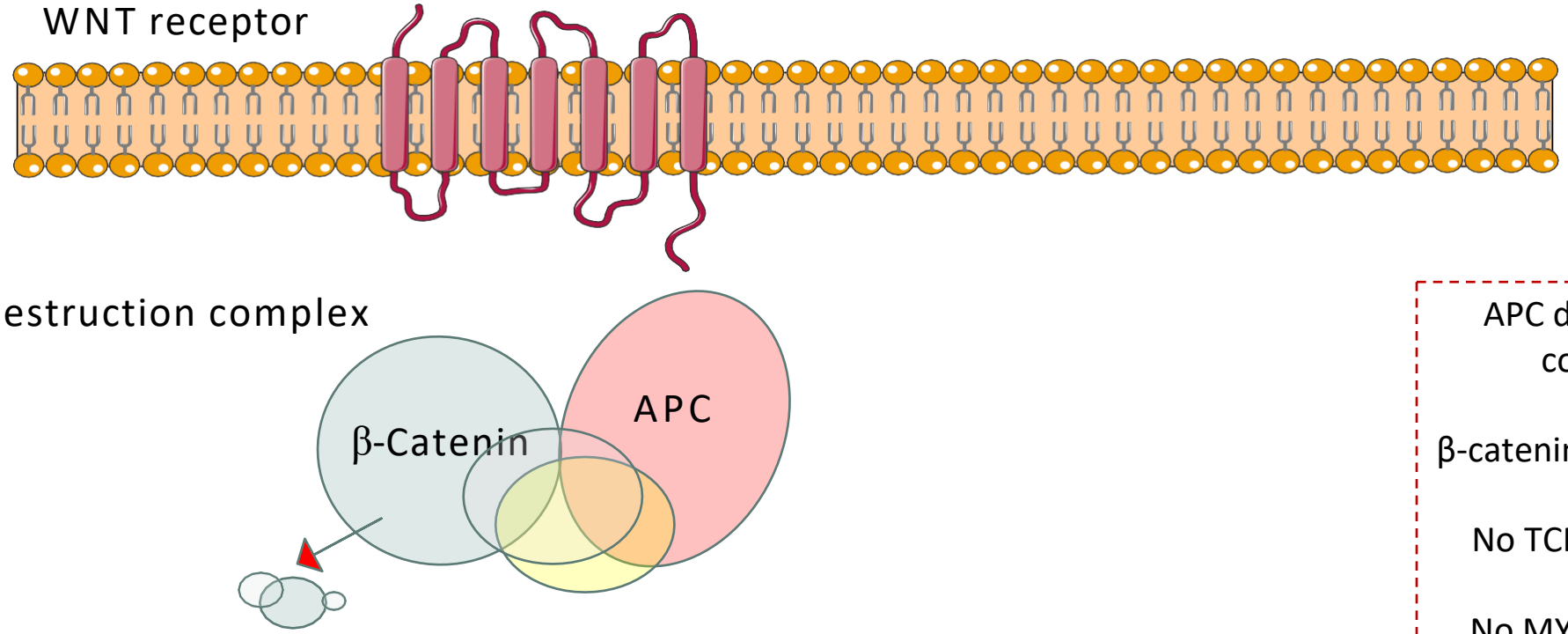
FAMILIAL ADENOMATOUS POLYPOSIS (FAP)

- *General*
  - Caused by a germline *APC* mutation
  - Autosomal dominant inheritance
  - Characterized by ≥100 conventional adenomas
  - Nearly 100% lifetime risk of CRC if untreated
- *Clinical features*
  - Adenomas develop during adolescence
  - Mean age of CRC onset: ~39 years
  - 95% penetrance before age 50
- *Associated syndromes*
  - Gardner syndrome
  - Turcot syndrome



APC/B-CATENIN PATHWAY

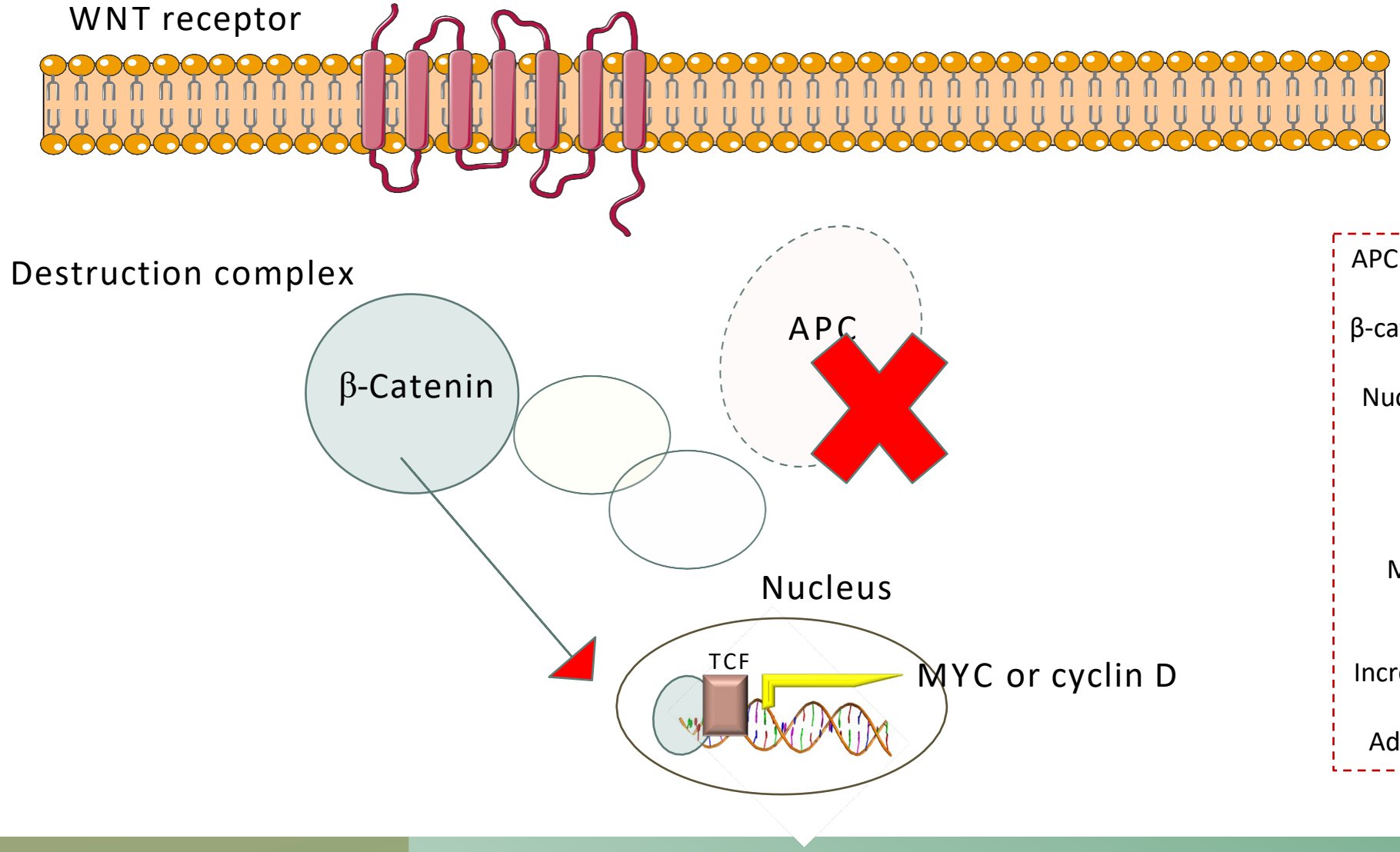
Resting



APC destruction complex
↓
β-catenin degradation
↓
No TCF activation
↓
No MYC / cyclin D transcription
↓
Controlled epithelial proliferation

APC/B-CATENIN PATHWAY

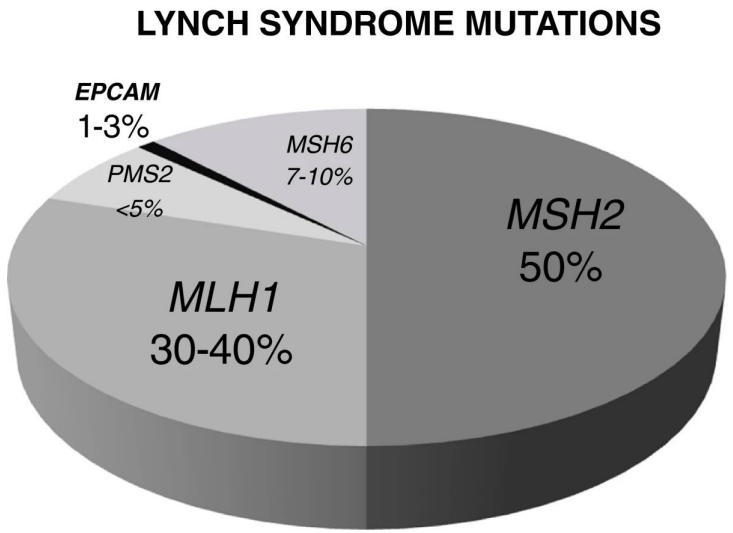
Tumor cells with mutated *APC*



APC germline mutation
↓
β-catenin accumulation
↓
Nuclear translocation of β-catenin
↓
TCF activation
↓
MYC and cyclin D expression
↓
Increased proliferation
↓
Adenoma formation

LYNCH SYNDROME

- *General*
  - Most common hereditary CRC syndrome
  - Most common monogenic cause of CRC
  - Autosomal dominant inheritance
  - Caused by a germline pathogenic variant in a DNA mismatch repair (MMR) gene (*MLH1, PMS2, MSH2, MSH6, EPCAM)*
  - Early onset (typically 40-50 years of age)
- *Clinical features*
  - Few or no precursor polyps
  - Predilection for proximal (right-sided colon)
  - Lifetime risk of CRC: ~70-80%
  - Mean age at CRC diagnosis: ~44 years
- *Associated malignancies*
  - Endometrial carcinoma (most common extracolonic malignancy)
  - Ovarian carcinoma
  - Upper urinary tract urothelial carcinoma
  - Gastric carcinoma
  - Small intestinal carcinoma

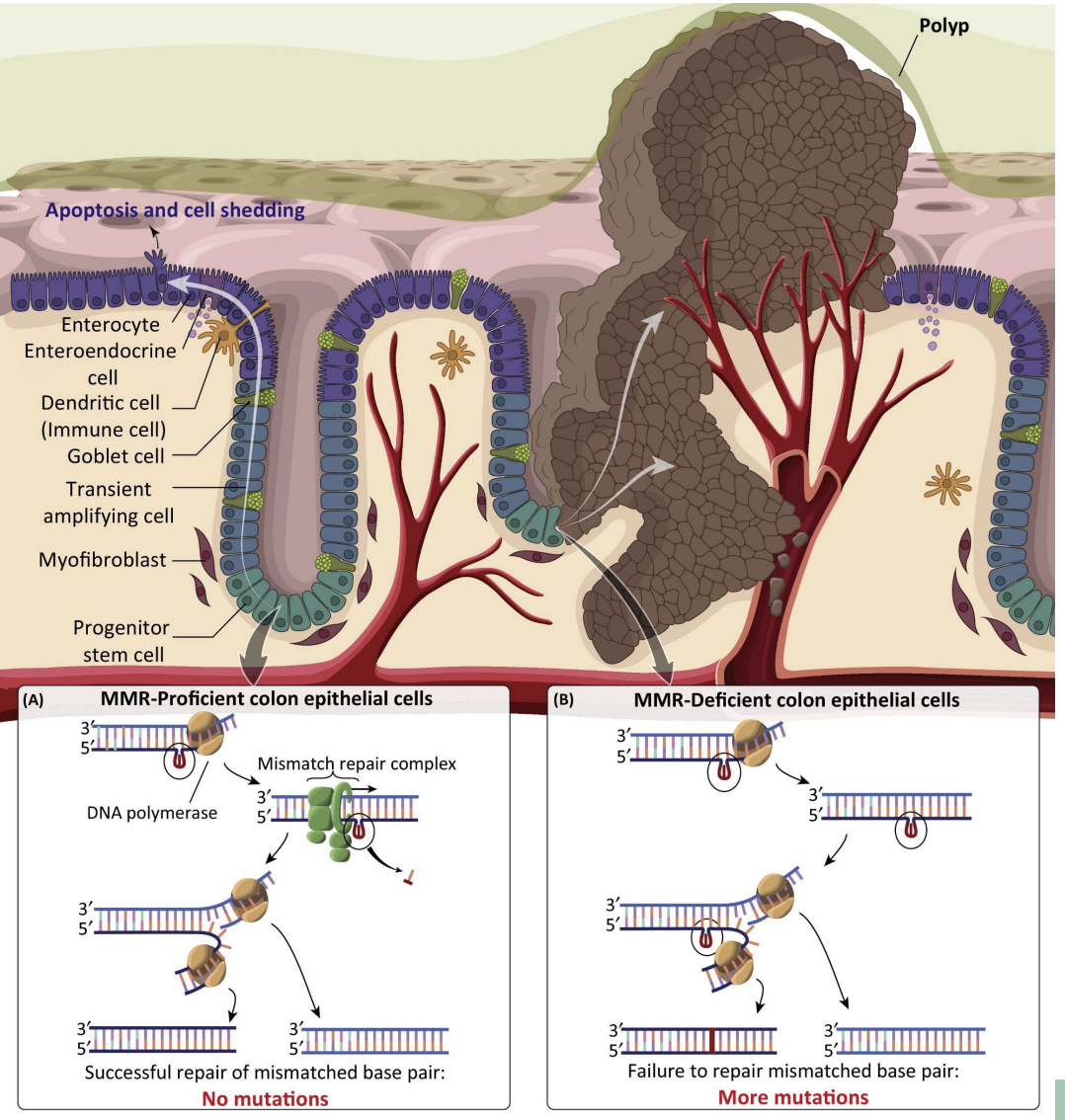


Hallmark of Lynch syndrome
Mismatch repair deficiency (dMMR)
↓
Microsatellite instability (MSI)

Lynch syndrome was previously called Hereditary Non-Polyposis Colorectal Cancer (HNPCC)

https://doi.org/10.1186/1897-4287-11-9

COLON CARCINOGENESIS- MISMATCH REPAIR(MMR) PATHWAY



Mutations in MMR genes

↓

Defect in MMR protein

↓

Accumulation of mutations

Accumulation of mutations in microsatellite DNA

↓

Microsatellite instability

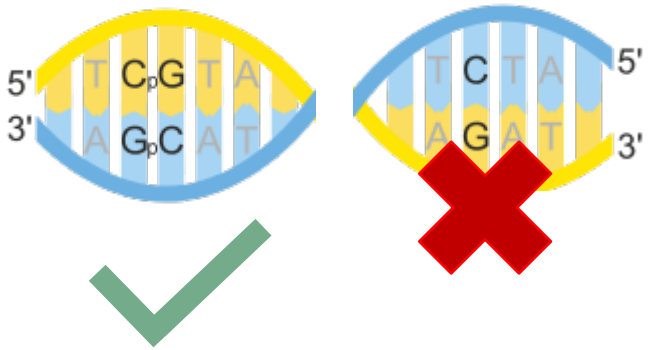
Microsatellite DNA:

- Short repeating DNA sequences (50,000 to 100,000 in human genome)
- Microsatellite genes generally are silent and located in the non-coding area.
- If these microsatellite genes are located in the promoter region of genes involved in cell survival and proliferation, cancer may develop
- Represent the most frequent sites of mutations in Lynch syndrome

https://link.springer.com/article/10.1007/s00535-019-01620-7#Fig2
https://www.cell.com/trends/molecular-medicine/fulltext/S1471-4914(16)00033-2

COLON CARCINOGENESIS-CPG ISLAND HYPERMETHYLATION

A CpG site, i.e., the " 5'—C—phosphate—G—3' " sequence of nucleotides, is indicated on one DNA strand (in yellow). On the reverse DNA strand (in blue), the complementary 5'—CpG—3' site is shown. A C-G base-paring between the two DNA strands is also indicated (right)



In humans, about 70% of promoters located near the transcription start site of a gene (proximal promoters) contain a CpG island.

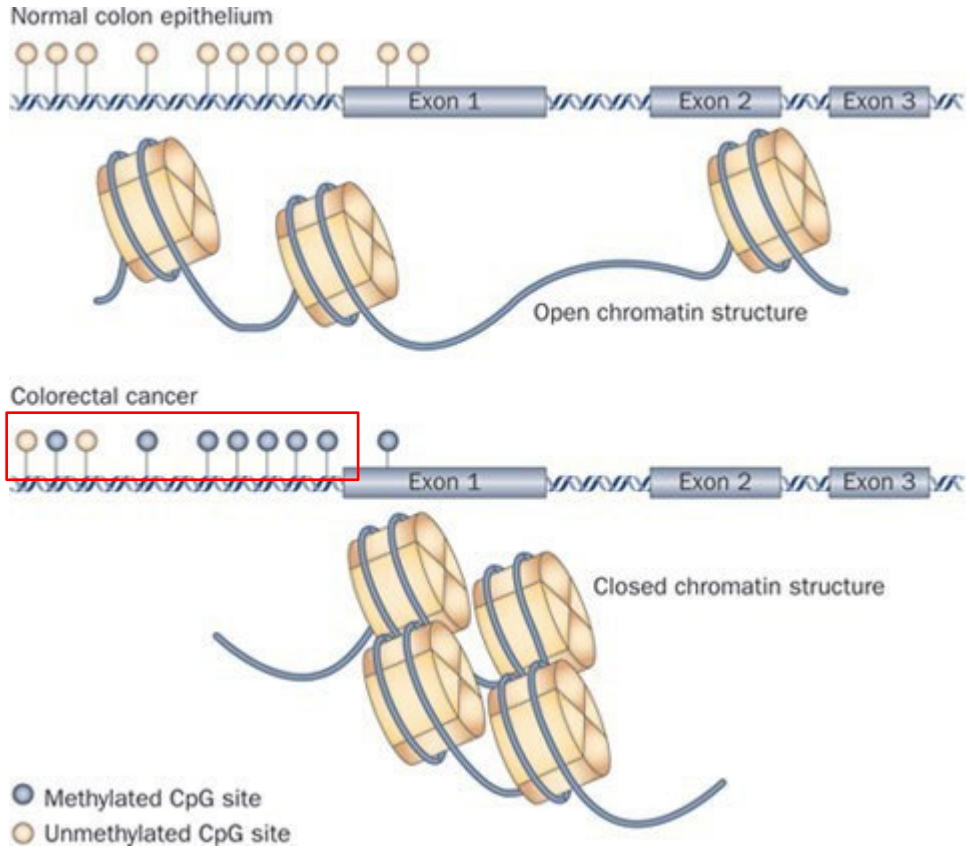
adenine (A), guanine (G) , thymine (T), cytosine (C)

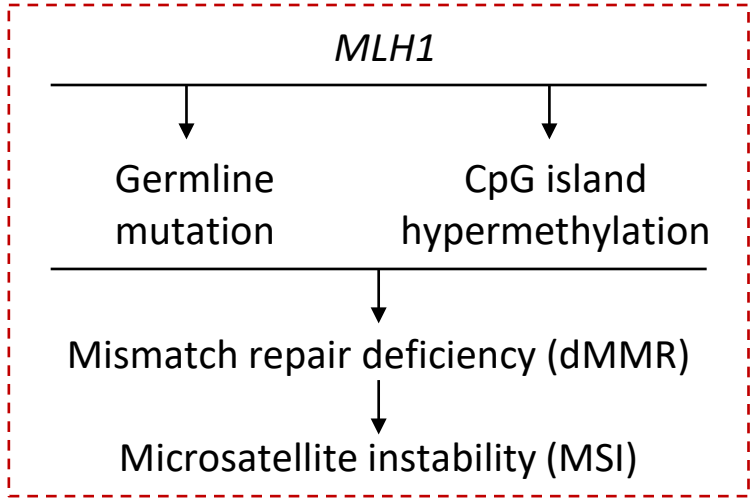
DNA methylation is an epigenetic mark that plays an essential role in ***regulating gene expression. CpG islands are DNA methylations regions*** in **promoters** known to regulate gene expression through transcriptional **silencing** of the corresponding gene. DNA methylation at CpG islands is crucial for gene expression and tissue-specific processes.

COLON CARCINOGENESIS-CPG ISLAND HYPERMETHYLATION

Normal

Neoplasia





No mutations in mismatch repair genes

However, CPG island hypermethylation in the *MLH1* promoter region will reduce MLH1 expression and repair function

https://www.researchgate.net/figure/CpG-island-DNA-hypermethylation-and-global-DNA-hypomethylation-in-colorectal-cancer-as_fig1_51727019


THE END